

The logic of arguments - Set 1 - Solutions

Specification Arg1-Arg4.

Answer key

Question	Answer	Topic
1	A	The logic of arguments
2	C	The logic of arguments
3	D	The logic of arguments
4	C	The logic of arguments
5	C	The logic of arguments
6	C	The logic of arguments
7	C	The logic of arguments
8	C	The logic of arguments
9	E	The logic of arguments
10	D	The logic of arguments
11	B	The logic of arguments
12	A	The logic of arguments

Worked solutions

Question 1 LGC-01 · The logic of arguments · spec Arg1

Consider the statement: **if n is a multiple of 6, then n is a multiple of 3**. Which of the following is its contrapositive?

Answer: A

The contrapositive of 'if A then B ' is 'if not B then not A '. It is logically equivalent to the original; B is the converse and C is the inverse.

Full method - how and why

The four relatives of an implication. Given $A \Rightarrow B$, you must be able to write the converse, inverse and contrapositive without hesitating.

- 1 Name the parts: A is ' n is a multiple of 6', B is ' n is a multiple of 3'.
why: Always label before manipulating - most errors here are bookkeeping errors.
- 2 The contrapositive is 'if not B then not A ': **if n is not a multiple of 3, then n is not a multiple of 6**.
why: Negate both parts **and** swap them. Doing only one of the two gives the wrong relative.
- 3 Sanity-check the meaning: if 3 does not divide n , then 6 certainly cannot, since every multiple of 6 is a multiple of 3.
why: The contrapositive is logically **equivalent** to the original, so it must also be true - a free check.

Where it goes wrong

- Choosing 'if n is a multiple of 3 then n is a multiple of 6' - that is the **converse** (swap only), and it is false ($n = 9$).
- Choosing 'if n is not a multiple of 6 then n is not a multiple of 3' - the **inverse** (negate only), also false ($n = 9$).

Takeaway Converse: swap. Inverse: negate. Contrapositive: swap **and** negate - and only the contrapositive is equivalent to the original.

Question 2 LGC-02 · The logic of arguments · spec Arg2

Let P be the statement ' $x > 2$ ' and Q the statement ' $x^2 > 4$ ', where x is a real number. Which one of the following is true?

Answer: C

$x > 2 \Rightarrow x^2 > 4$, so P is sufficient for Q . But $x = -3$ gives $x^2 > 4$ without $x > 2$, so P is not necessary.

Full method - how and why

Necessary/sufficient questions are two separate yes-no checks: does $P \Rightarrow Q$? does $Q \Rightarrow P$?

- 1 Test sufficiency ($P \Rightarrow Q$): if $x > 2$ then $x^2 > 4$. **True**, so P is sufficient.
why: Squaring preserves the inequality for positive numbers.
- 2 Test necessity ($Q \Rightarrow P$): try to break it. $x = -3$ gives $x^2 = 9 > 4$ but $x \not> 2$.
why: One counterexample is enough to destroy an implication; negative numbers are the obvious place to look when squaring.
- 3 So P is sufficient but not necessary.
why: Sufficient = 'enough to guarantee'; necessary = 'can't do without'.

Where it goes wrong

- Getting the direction backwards: ' P is necessary for Q ' means $Q \Rightarrow P$, not $P \Rightarrow Q$.
- Only checking positive x and concluding the two are equivalent.

Takeaway P sufficient for Q means $P \Rightarrow Q$; P necessary for Q means $Q \Rightarrow P$. Check each direction separately, hunting for counterexamples.

Question 3 LGC-03 · The logic of arguments · spec Arg4

Which of the following is the negation of the statement '**every student in the class scored more than 50**'?

Answer: D

The negation of 'for all x , $P(x)$ ' is 'there exists x such that not $P(x)$ '. 'Not more than 50' means '50 or less'.

Full method - how and why

Negating a universal statement. 'All' flips to 'at least one', and the property inside flips too.

- 1 Write it formally: 'for all students s , $\text{mark}(s) > 50$ '.

why: Formalising stops the English from misleading you.

- 2 Negate: 'there exists a student s with $\text{mark}(s) > 50$ '.

why: $\neg \forall x P(x) \equiv \exists x \neg P(x)$; the quantifier flips and the inside is negated.

- 3 Translate > 50 into ' ≤ 50 ', i.e. **50 or less**.

why: The negation of 'greater than' is 'less than **or equal to**' - the boundary belongs to the negation.

- 4 So: at least one student scored 50 or less.

why: One failure is all it takes to make 'every' false.

Where it goes wrong

- Over-negating to 'every student scored 50 or less' - that is far stronger than the negation and is not implied.
- Writing 'less than 50' and losing the case of exactly 50.

Takeaway Negating 'all ... are' gives 'at least one ... is not' - and remember $\neg(x > a)$ is $x \leq a$.

Question 4 LGC-04 · The logic of arguments · spec Arg1

Consider the statement about a real number x : **if $x^2 = x$, then $x = 1$** . Which one of the following is true?

Answer: C

$x = 0$ satisfies $x^2 = x$ but not $x = 1$, so the statement is false. The converse, 'if $x = 1$ then $x^2 = x$ ', is true.

Full method - how and why

Judge a statement and its converse separately. Each needs either a proof or a counterexample.

- 1 The statement: if $x^2 = x$ then $x = 1$. Solve $x^2 = x$: $x(x - 1) = 0$, so $x = 0$ or $x = 1$.

why: Never divide by x - that is exactly how the root $x = 0$ gets lost.

- 2 $x = 0$ satisfies the hypothesis but not the conclusion, so the statement is **false**.

why: A single counterexample settles it.

- 3 The converse: if $x = 1$ then $x^2 = x$. Substituting gives $1 = 1$. **True**.

why: Verifying one value is enough because the hypothesis pins x down completely.

- 4 So: false statement, true converse.

why: Match to the option list.

Where it goes wrong

- Dividing $x^2 = x$ by x , getting only $x = 1$, and wrongly declaring the statement true.
- Assuming a false statement must have a false converse - the two are independent.

Takeaway A statement and its converse have no logical connection; test each on its own, and never divide an equation by a variable.

Question 5 LGC-07 · The logic of arguments · spec Arg2

Let P : 'the quadrilateral $ABCD$ is a square' and Q : 'the quadrilateral $ABCD$ has four equal sides'. Which one of the following is true?

Answer: C

A square has four equal sides ($P \Rightarrow Q$, sufficient), but a rhombus that is not a square shows Q does not imply P (not necessary).

Full method - how and why

The same two-direction test, now in geometry. Look for a shape satisfying one condition but not the other.

1 Sufficiency: every square has four equal sides, so $P \Rightarrow Q$. **True.**

why: Straight from the definition of a square.

2 Necessity: does four equal sides force a square? A rhombus with angles 60° and 120° has four equal sides and is not a square.

why: A square also needs right angles; equal sides alone do not fix the angles.

3 So P is sufficient but not necessary for Q .

why: P is the stronger condition, so it implies the weaker one but not conversely.

Where it goes wrong

- Confusing which is the stronger condition: the more demanding property (square) is sufficient for the less demanding one (four equal sides).
- Failing to produce a concrete counterexample and guessing.

Takeaway A stronger condition is sufficient for a weaker one; the weaker is necessary for the stronger. Name a counterexample to confirm.

Question 6 LGC-08 · The logic of arguments · spec Arg4

Which of the following is the negation of 'there exists a positive integer n such that $n^2 + n + 41$ is not prime'?

Answer: C

Negating 'there exists n with not $P(n)$ ' gives 'for all n , $P(n)$ '.

Full method - how and why

Negating an existential statement whose inside is already a negation - so two flips happen.

1 Formalise: ' $\exists n$ such that $n^2 + n + 41$ is **not** prime'.

why: Note the inner 'not'.

2 Negate the quantifier: $\neg \exists n P(n) \equiv \forall n \neg P(n)$.

why: 'There is one' becomes 'there is none', i.e. 'every one fails'.

3 Negate the inside: 'not (not prime)' = 'prime'.

why: Double negation cancels.

4 Result: **for every positive integer n , $n^2 + n + 41$ is prime.**

why: Exactly one of the original and this is true; in fact this one is false, since $n = 41$ gives $41^2 + 41 + 41 = 41 \times 43$.

Where it goes wrong

- Leaving the inner 'not' in place: 'for every n , $n^2 + n + 41$ is not prime' - that negates the quantifier but not the property.
- Keeping 'there exists' and only flipping the property.

Takeaway $\neg \exists x \neg P(x) \equiv \forall x P(x)$: flip the quantifier **and** the statement inside.

Question 7 LGC-10 · The logic of arguments · spec Arg2

For real numbers a and b , let P : ' $a = b$ ' and Q : ' $a^2 = b^2$ '. Which one of the following is true?

Answer: C

$a = b \Rightarrow a^2 = b^2$ (sufficient), but $a = 1, b = -1$ gives $a^2 = b^2$ without $a = b$ (not necessary).

Full method - how and why

The squaring trap again, in its purest form.

- 1 Sufficiency: if $a = b$ then squaring both sides gives $a^2 = b^2$. **True.**
why: Doing the same thing to equal quantities keeps them equal.
- 2 Necessity: does $a^2 = b^2 \Rightarrow a = b$? Take $a = 1, b = -1$: $1 = 1$ but $a \neq b$.
why: $a^2 = b^2$ only gives $a = \pm b$.
- 3 So P is sufficient but not necessary.
why: Squaring loses sign information, so it cannot be reversed.

Where it goes wrong

- Treating squaring as reversible - it is exactly this that makes squaring an equation introduce extra solutions.
- Only trying positive values.

Takeaway $a = b \Rightarrow a^2 = b^2$, but $a^2 = b^2 \Rightarrow a = \pm b$. Squaring is a one-way street.

Question 8 LGC-11 · The logic of arguments · spec Arg1

The statement 'if a triangle is equilateral then it is isosceles' is true. Which of the following must therefore also be true?

- I If a triangle is isosceles then it is equilateral.
- II If a triangle is not isosceles then it is not equilateral.
- III If a triangle is not equilateral then it is not isosceles.

Answer: C

Only the contrapositive (II) is logically equivalent to the original. I is the converse and III is the inverse; both are in fact false here.

Full method - how and why

Given a true implication, only one of the three relatives is guaranteed. Identify which of I, II, III is the contrapositive.

- 1 I is 'isosceles $\Rightarrow$ equilateral' - the **converse**. Not guaranteed, and here false (a 3, 3, 5 triangle).
why: Swapped but not negated.
- 2 II is 'not isosceles $\Rightarrow$ not equilateral' - the **contrapositive**. Always guaranteed. **True.**
why: Swapped and negated; logically equivalent to the original.
- 3 III is 'not equilateral $\Rightarrow$ not isosceles' - the **inverse**. Not guaranteed, and here false (the same 3, 3, 5 triangle).
why: Negated but not swapped - and the inverse is just the contrapositive of the converse, so it carries the converse's truth value, not the original's.
- 4 So II only.
why: Match to the list.

Where it goes wrong

- Accepting III because both its parts are negated - negation alone is not enough; the parts must also swap.
- Accepting I because equilateral triangles are also isosceles - that supports the original, not the converse.

Takeaway From $A \Rightarrow B$ you get exactly one free statement: $\neg B \Rightarrow \neg A$.

Question 9 LGC-13 · The logic of arguments · spec Arg4

Which of the following is the negation of ' $x > 3$ and $y \leq 5$ '?

Answer: E

Not $(A \text{ and } B)$ is $(\text{not } A) \text{ or } (\text{not } B)$. Not $x > 3$ is $x \leq 3$; not $y \leq 5$ is $y > 5$.

Full method - how and why

De Morgan's law: negating an 'and' turns it into an 'or', with both parts negated.

1 $\neg(A \text{ and } B) \equiv (\neg A) \text{ or } (\neg B)$.

why: For the conjunction to fail, it is enough that **one** part fails.

2 Negate the first part: $\neg(x > 3)$ is $x \leq 3$.

why: The boundary $x = 3$ satisfies the negation, since $3 > 3$ is false.

3 Negate the second: $\neg(y \leq 5)$ is $y > 5$.

why: Strict inequality, because $y = 5$ satisfied the original.

4 Combine with 'or': $x \leq 3$ **or** $y > 5$.

why: The joining word flips too - this is the half most often forgotten.

Where it goes wrong

- Keeping 'and': ' $x \leq 3$ and $y > 5$ ' is far too strong - it demands both failures.
- Getting the boundaries wrong: writing $x < 3$ or $y \geq 5$, which mishandles $x = 3$ and $y = 5$.

Takeaway De Morgan: $\text{not}(A \text{ and } B) = \text{not}A \text{ or } \text{not}B$; $\text{not}(A \text{ or } B) = \text{not}A \text{ and } \text{not}B$. Negate the connective as well as the parts.

Question 10 LGC-14 · The logic of arguments · spec Arg1

For a positive integer n , consider the statement: **if n is prime, then n is odd**. Which one of the following is true?

Answer: D

$n = 2$ is an even prime, so the statement is false. The converse, 'if n is odd then n is prime', fails for $n = 9$.

Full method - how and why

Statement and converse again, with the classic $n = 2$ counterexample.

1 Statement: primes are odd? $n = 2$ is prime and even. **False**.

why: 2 is the only even prime - it is the standard counterexample and worth memorising.

2 Converse: odd numbers are prime? $n = 9 = 3 \times 3$ is odd and composite. **False**.

why: Also $n = 1$, which is odd and not prime by definition.

3 So both are false.

why: Match to the option list.

Where it goes wrong

- Forgetting that 2 is prime and declaring the statement true.
- Forgetting that 1 is **not** prime when hunting counterexamples.

Takeaway 2 is the even prime and 1 is not prime - the two facts behind most prime-related counterexamples.

Question 11 LGC-17 · The logic of arguments · spec Arg1

Let x be a real number. Which of the following statements is equivalent to ' $x \geq 2$ if and only if $x^3 \geq 8$ '?

Answer: B

'If and only if' is the conjunction of the implication and its converse.

Full method - how and why

Unpacking 'if and only if' into its two implications.

- 1 'A if and only if B' means both 'A if B' ($B \Rightarrow A$) and 'A only if B' ($A \Rightarrow B$).

why: The phrase is literally built from 'if' and 'only if' - each supplies one direction.

- 2 With $A: x \geq 2$ and $B: x^3 \geq 8$, that is $(x \geq 2 \Rightarrow x^3 \geq 8)$ **and** $(x^3 \geq 8 \Rightarrow x \geq 2)$.

why: The conjunction of the implication and its converse.

- 3 (Both happen to be true here, since cubing is increasing - but the question only asks for the equivalent form.)

why: Read what is asked: the structure, not the truth value.

Where it goes wrong

- Giving only one implication - that is half the statement.
- Joining the two with 'or' instead of 'and'.

Takeaway $A \Leftrightarrow B$ is exactly $(A \Rightarrow B)$ **and** $(B \Rightarrow A)$.

Question 12 LGC-19 · The logic of arguments · spec Arg2

Let f be a function defined for all real numbers. $P: 'f'(a) = 0$ '; $Q: 'f$ has a stationary point at $x = a$ '. Which one of the following is true?

Answer: A

A stationary point is by definition a point where the derivative is zero, so P and Q are equivalent.

Full method - how and why

Sometimes the two statements are the same by **definition** - recognise that and both directions are immediate.

- 1 Recall the definition: f has a stationary point at $x = a$ precisely when $f'(a) = 0$.

why: That is what 'stationary' means - the rate of change is zero.

- 2 So $P \Rightarrow Q$ and $Q \Rightarrow P$ both hold trivially.

why: Definitions are biconditional.

- 3 Hence P is necessary and sufficient for Q .

why: Match to the option list.

Where it goes wrong

- Confusing this with 'turning point': $f'(a) = 0$ does **not** guarantee a maximum or minimum ($y = x^3$ at $x = 0$). But the question says stationary point, not turning point.
- Worrying about functions that are not differentiable - the question states f is defined and $f'(a)$ is referred to, so differentiability is assumed.

Takeaway A definition gives both implications at once; but stationary $\neq$ turning - the extra condition is a **sign change** in f' .